```
def to_8bit(num):
    return format(num & 0xFF, '08b')


def binary_addition(a, b):
    binary_a = to_8bit(a)
    binary_b = to_8bit(b)

    result = (a + b) & 0xFF

    # Overflow occurs when operands have the same sign
    # but the result has a different sign
    overflow = (a >= 0 and b >= 0 and result >= 128) or \
               (a < 0 and b < 0 and result < 128)

    if result >= 128:
        decimal_result = result - 256
    else:
        decimal_result = result

    print("Operand A:", binary_a)
    print("Operand B:", binary_b)

    print("Binary Result:", to_8bit(result))
    print("Decimal Result:", decimal_result)

    if overflow:
        print("Arithmetic Overflow: Yes")
    else:
        print("Arithmetic Overflow: No")


print("Expression (a): 25 + 18")
binary_addition(25, 18)

print("\nExpression (b): -25 + -18")
binary_addition(-25, -18)
```

//Output//

Expression (a): 25 + 18
Operand A: 00011001
Operand B: 00010010
Binary Result: 00101011
Decimal Result: 43
Arithmetic Overflow: No

Expression (b): -25 + -18
Operand A: 11100111
Operand B: 11101110
Binary Result: 11010101
Decimal Result: -43
Arithmetic Overflow: No